

Wave Fetch Analysis

Group 3

Hamidreza Khalaj Zahraei (300204093)

Ali Özuysal (310204059)

1 Introduction

This report presents the fetch analysis for a coastal site on the Black Sea following the method taught in class. Fetch is the unobstructed distance over which wind generates waves. The analysis uses $\pm 7.5^\circ$ sectors around each 22.5° main direction, with cosine-squared weighting to compute effective fetch lengths:

$$F_{\text{eff}} = \frac{\sum F_i \cos^2 \alpha_i}{\sum \cos \alpha_i},$$

where α_i is the angular deviation of a measured ray from the main direction.

2 Methodology

1. **Origin:** The starting point was moved 1–2 km offshore from the original land point (coordinates: 41.934003°N, 28.069026°E) to avoid local sheltering.
2. **Radial measurements:** For each of the 16 main directions ($0^\circ, 22.5^\circ, \dots, 337.5^\circ$), fetch distances were measured along three sub-directions: $\theta_{\text{main}} - 7.5^\circ$, θ_{main} , and $\theta_{\text{main}} + 7.5^\circ$. The intersection with the shoreline was recorded (latitude, longitude) using Google Earth. The Haversine formula then gave the distance from the origin. Figure 1 shows the 48 radials over the Black Sea basin.
3. **Effective fetch calculation:** The above cosine-squared formula was applied to each sector.
4. **Threshold:** Any main direction with $F_{\text{eff}} < 5$ km was excluded.

3 Results

Table 1 lists the effective fetch lengths for all 16 directions. Figure 2 shows the fetch rose – a polar bar plot of the effective distances. The red dashed circle indicates the 5 km threshold.

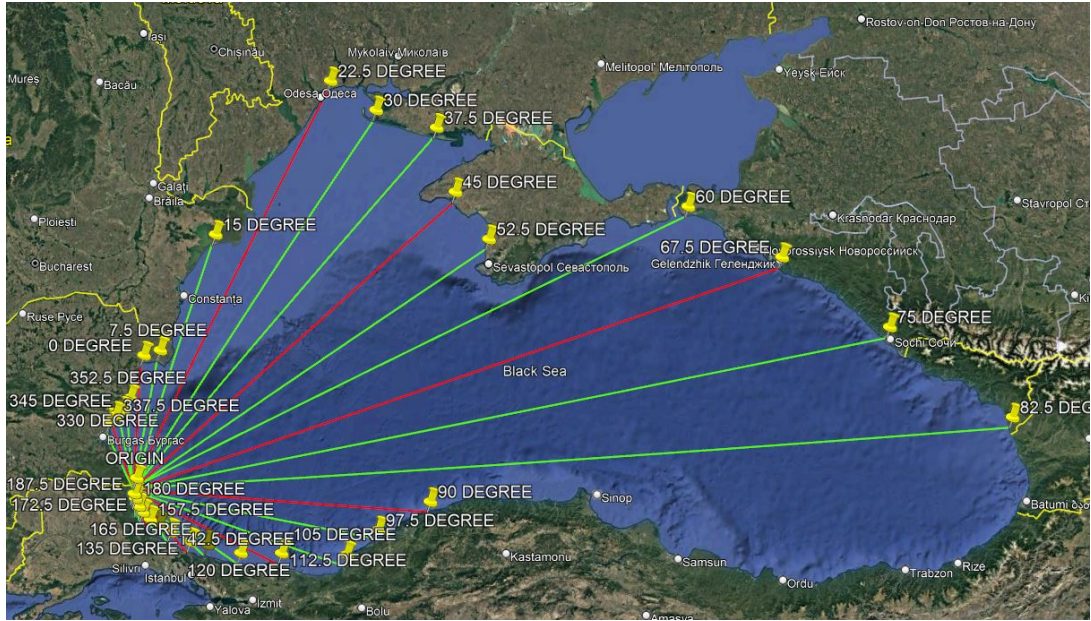


Figure 1: Map of the Black Sea showing the 48 radials (3 per 22.5° sector) from the offshore origin. The annotated bearings indicate the main directions.

Most critical three directions

Based on the largest effective fetch lengths, the three most critical directions for wave generation are:

1. **67.5°** – 1074.12 km
2. **90.0°** – 597.69 km
3. **45.0°** – 554.31 km

4 Conclusion

The site is exposed to waves from most directions, with the longest effective fetch from the east-northeast (67.5°). Directions from 202.5° to 315° are fully sheltered (fetch <5 km). These effective fetch lengths will be used as input to the JONSWAP wave growth equations for design wave height and period.

Attachments

The Python code used for the calculation and the terminal output are submitted as separate files (HW2_GroupX_code.py and HW2_GroupX_output.txt). The figures (1.jpg and fetch_rose.png) are included.

Table 1: Effective fetch lengths for each 22.5° direction.

Bearing (°)	Effective Fetch (km)	Status
0.0	140.68	VALID
22.5	478.25	VALID
45.0	554.31	VALID
67.5	1074.12	VALID
90.0	597.69	VALID
112.5	219.42	VALID
135.0	103.06	VALID
157.5	54.55	VALID
180.0	27.11	VALID
202.5	0.00	EXCLUDED
225.0	0.00	EXCLUDED
247.5	0.00	EXCLUDED
270.0	0.00	EXCLUDED
292.5	0.00	EXCLUDED
315.0	0.00	EXCLUDED
337.5	83.25	VALID

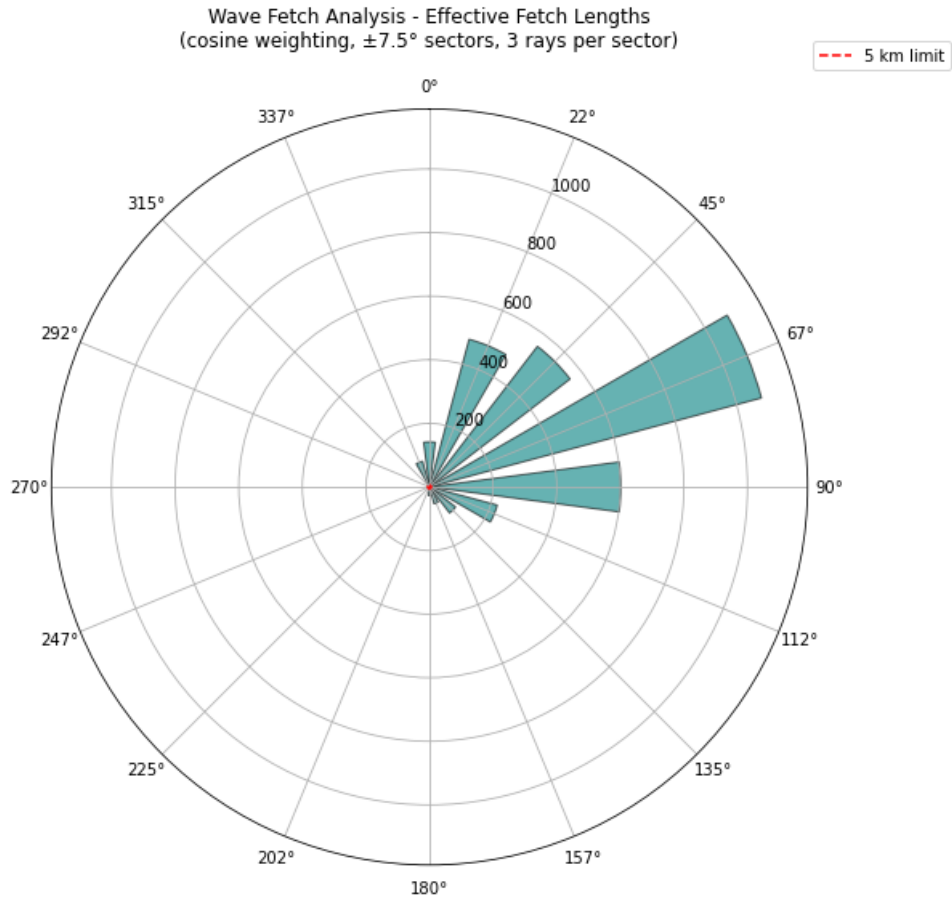


Figure 2: Fetch rose showing effective fetch lengths (km). The red dashed circle marks the 5 km threshold.